

PAUL PIOTROWSKI



Miami, FL | (786) 236-5348 | paulpiotrowski@gmail.com | linkedin.com/in/paul-pio | github.com/PaulPio | paulpio.com

EDUCATION

B.S. Computer Science — Florida International University

May 2025 – Apr 2028

GPA: 3.76 | Clubs: INIT, Google Developer Group, SHPE, IEEE, Code Crunch, Panther Robotics, AAVE, CASHI

TECHNICAL SKILLS

Languages & Frontend: TypeScript, JavaScript, Python, C++, Java, React 19, Next.js, Tailwind CSS, Vite

Backend, Data & AI: Node.js, REST APIs, Supabase, MongoDB Atlas, Gemini, Gemma, Claude, Mistral AI, Hermes

Tools & Infra: Git, GitHub Actions, GitLab, Vercel, Docker, Puppeteer/Playwright, Excel (Power Query, VLOOKUP, Pivot Tables)

WORK EXPERIENCE

Undergraduate Research Assistant — Florida International University, Miami, FL

Aug 2026 – Present

Stack: C# · Unity · Mixamo MCP · Local LLMs · Python · Apple Vision Pro · XR/AR

- Develop and test C# and Unity modules for an FIU Police Department XR/AR training program on the Apple Vision Pro.
- Created character animations in Unity through the Mixamo MCP pipeline, expanding the trainee scenarios the simulation supports.
- Integrated local LLM calls into the XR/AR simulation so the virtual character reacts to each trainee in real time.
- Wrote Python logging scripts that capture structured training session data and improved the trainer review dashboard.

Software Engineer Intern — City of Coral Gables, Coral Gables, FL

May – Jun 2026

Stack: Python · Gemini Veo · PyGame · OpenCV · Playwright · Raspberry Pi

- Engineered an AI smart city kiosk proof of concept on a Raspberry Pi 5 in 3 weeks, delivering an interactive IoT installation.
- Programmed a state-driven Pygame frontend to manage the AI avatar's lifecycle, triggering Gemini Veo-generated video states (Idle, Listening, Speaking) in real time.

Stride Rent a Car — Miami, FL

Sep 2022 – May 2026

Software Engineer

Dec 2025 – May 2026

Stack: React · TypeScript · Tailwind CSS · Supabase · Rently API · HQ Car Rental API · Vercel

- Architected a 6-module enterprise dashboard integrating the Rently and HQ Car Rental APIs to manage over 1,200 vehicles and 5,000 monthly reservations, saving over 45 hours of manual work weekly.
- Developed DisputeDefender, an AI tool that extracts rental agreement data from PDFs and generates bank-ready rebuttal letters, cutting chargeback response time 90% from 30 minutes to under 3.
- Shipped every feature from requirement to production as the sole engineer across reservations, fleet availability, and chargeback systems for a company processing over 5,000 reservations monthly.

Sales Supervisor

Sep 2022 – Dec 2025

- Supervised sales staff while processing customer reservations and resolving complaints for a fleet of over 1,200 vehicles.
- Handled damage claims with insurers and wrote chargeback rebuttals to banks, a process later automated by DisputeDefender.
- Compiled monthly sales and insurance reports in Excel using VLOOKUP, Power Query, pivot tables, and charts.

PROJECTS

UtiliTies — 2nd Place, Gridlock Challenge, ShellHacks 2026

Sep 2026

Live: github.com/AlexWS12/shellhacks26

Stack: Python · FastAPI · Gemini · Jev · Next.js · PostgreSQL · MapLibre

- Placed 2nd at ShellHacks, Florida's largest hackathon with over 1,400 attendees, by mapping over 250 real utility projects.
- Developed a multi-agent pipeline with Gemini and Jev that detects cost saving and cooperation opportunities between utilities.

ResumeFit — 1st Place, 305 SummerCodex (Build with Gemma)

Jul 2026

Live: resume-fit-project.vercel.app

Stack: Next.js 16 · TypeScript · Gemma (OpenRouter) · Zod · Supabase Auth · pdf-lib · unpdf · Vercel

- Won 1st place solo in the Opportunity & Access track by building an AI tool that closes the resume loop end to end.
- Engineered a Gemma pipeline that health-checks resumes and ranks them against hundreds of real SWE internships.

Home IoT System — ESP32 & Raspberry Pi

Jun 2026 – Present

Live: github.com/PaulPio/IOT

Stack: ESP32 · Raspberry Pi · SQLite · Python · Telegram Bot API · Pi-hole · Tailscale · Hermes

- Networked ESP32 microcontrollers to a Raspberry Pi hub that logs temperature and humidity history to a SQLite dashboard.
- Designed a motion sensing doorbell that sends real time alerts via a custom Telegram bot, with remote access over Tailscale.
- Deployed Pi-hole as a network-wide DNS ad blocker and routed devices through Tailscale to block ads from anywhere.
- Configured a personal AI agent on the Hermes OS project, with specialized agents dedicated to separate tasks.

EXTRACURRICULAR ACTIVITIES

Research Ambassador — Florida International University, Miami, FL

Sep 2026 – Present

- Help FIU students discover and pursue undergraduate research opportunities across campus.